



DSTL (BCS-303) — Unit 5

Previous Year Questions with Step-by-Step Solutions

ABES Engineering College
Prepared for Practice and Revision

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Contents

1	III Semester Theory Examination 2017–18	2
2	III Semester Theory Examination 2019–20	4
3	III Semester Theory Examination 2020–21	7
4	III Semester Theory Examination 2021–22	8

How to Use

Each PYQ is followed by a line-by-line solution. For Hasse diagrams, we explain the construction steps (levels/covers). Redraw neatly in your answer booklet.

1 III Semester Theory Examination 2017–18

2017–18, Section A, Q1

Define Eulerian path, Eulerian circuit and Eulerian graph.

Solution

Eulerian path: In a connected graph G , an *Eulerian path* is a walk which traverses every edge of G *exactly once*, but the starting and ending vertices may be different.

Eulerian circuit (Eulerian cycle): An *Eulerian circuit* is an Eulerian path whose starting and ending vertices are the same.

Eulerian graph: A connected graph G is called *Eulerian* if it contains an Eulerian circuit, i.e. there exists a closed walk that uses every edge of G exactly once.

Characterization (for revision): A connected graph is Eulerian if and only if every vertex of the graph has even degree.

2017–18, Section A, Q2

Explain edge coloring and k -edge coloring.

Solution

Edge coloring: Let $G = (V, E)$ be a graph. An *edge coloring* of G is an assignment of colors to the edges so that no two adjacent edges (edges having a common end vertex) receive the same color.

k -edge coloring: If the colors used come from a set of exactly k colors, then the edge coloring is called a *k -edge coloring*.

The minimum number k for which G admits a k -edge coloring is called the *edge chromatic number* or *chromatic index* of G and is usually denoted by $\chi'(G)$.

2017–18, Section A, Q3

Define chromatic number and isomorphic graph.

Solution

Chromatic number: A *proper vertex coloring* of a graph G is an assignment of colors to each vertex such that no two adjacent vertices have the same color. The *chromatic number* $\chi(G)$ of G is the minimum number of colors needed for a proper vertex coloring of G .

Isomorphic graphs: Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are called *isomorphic* if there exists a bijection

$$\varphi : V_1 \rightarrow V_2$$

such that $\{u, v\} \in E_1$ if and only if $\{\varphi(u), \varphi(v)\} \in E_2$. Intuitively, isomorphic graphs have the same structure; only vertex labels differ.

2017–18, Section B, Q1

Define the following with one example each:

1. Bipartite graph.

2. Complete graph.
3. How many edges are there in K_7 and $K_{3,6}$?
4. Planar graph.

Solution

(i) Bipartite graph: A graph $G = (V, E)$ is *bipartite* if its vertex set can be partitioned into two disjoint sets V_1 and V_2 such that every edge of G has one end in V_1 and the other end in V_2 ; i.e. no edge joins two vertices of the same part.

Example: A simple even cycle such as C_4 or the complete bipartite graph $K_{2,3}$.

(ii) Complete graph: A *complete graph* on n vertices, denoted by K_n , is a simple graph in which every pair of distinct vertices is joined by exactly one edge.

Example: K_4 has 4 vertices and every vertex is adjacent to all other 3 vertices.

(iii) Number of edges in K_7 and $K_{3,6}$:

For a complete graph K_n , the number of edges is

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}.$$

So

$$|E(K_7)| = \binom{7}{2} = \frac{7 \cdot 6}{2} = 21.$$

For a complete bipartite graph $K_{m,n}$, the number of edges is

$$|E(K_{m,n})| = mn.$$

Thus

$$|E(K_{3,6})| = 3 \cdot 6 = 18.$$

(iv) Planar graph: A graph G is called *planar* if it can be drawn in the plane without any edges crossing (except at their end vertices). Such a drawing is called a *plane embedding* or *planar representation* of G .

2017–18, Section C, Q1

Define and explain any two of the following:

1. Euler graph.
2. Adjacency matrix of a graph.

Solution

Euler graph: A connected graph is called an *Euler graph* (or Eulerian graph) if it has an Eulerian circuit, i.e. there exists a closed walk that traverses every edge exactly once. Equivalently, a connected graph is Eulerian if and only if all vertices have even degree.

Adjacency matrix: Let G be a simple graph on vertex set $V = \{v_1, v_2, \dots, v_n\}$. The *adjacency matrix* of G is the $n \times n$ matrix $A = (a_{ij})$ where

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge between } v_i \text{ and } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

For a simple undirected graph the adjacency matrix is symmetric and has all diagonal entries 0. For example, the adjacency matrix of the path $v_1 - v_2 - v_3$ is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

2 III Semester Theory Examination 2019–20

2019–20, Section A, Q1

Show that there does not exist a simple graph with 5 vertices having degrees 1, 3, 4, 2, 3 respectively.

Solution

Let the degrees of the 5 vertices be

$$d_1 = 1, d_2 = 3, d_3 = 4, d_4 = 2, d_5 = 3.$$

First, by the **Handshaking Lemma**, we know that in any simple graph

$$\sum_{i=1}^5 d_i = 2|E|.$$

Compute the sum:

$$1 + 3 + 4 + 2 + 3 = 13.$$

But 13 is odd, whereas $2|E|$ is always even. Hence such a degree sequence cannot arise from any graph.

Therefore, there does not exist a simple graph with these vertex degrees.

2019–20, Section A, Q2

Define Pigeonhole Principle.

Solution

Simple Pigeonhole Principle: If $n + 1$ or more objects are placed into n boxes, then at least one box contains ≥ 2 objects.

Generalized Pigeonhole Principle: If N objects are placed into k boxes, then some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

2019–20, Section B, Q1

Define planar graph. Prove that for any connected planar graph

$$v - e + r = 2,$$

where v , e and r are the numbers of vertices, edges and regions (faces) of the planar drawing respectively.

Solution

Planar graph definition: already given earlier.

Euler's Formula: For any connected planar graph

$$v - e + r = 2.$$

Proof (by induction on the number of edges):

Base case: For a tree (a connected graph with no cycles) embedded in the plane, we have $e = v - 1$. It divides the plane into exactly one region (the unbounded exterior region), i.e. $r = 1$. So

$$v - e + r = v - (v - 1) + 1 = 2.$$

Induction step: Assume the formula holds for all connected planar graphs with $e - 1$ edges. Let G be a connected planar graph with e edges.

Since G is connected and has at least one cycle (otherwise it is a tree and already handled), choose an edge e_0 that lies on a cycle. Delete e_0 . The resulting graph G' is still connected and planar.

In the planar embedding, removing an edge on a cycle merges two adjacent regions into one, so

$$v' = v, \quad e' = e - 1, \quad r' = r - 1.$$

By the induction hypothesis applied to G' ,

$$v' - e' + r' = 2.$$

Substituting,

$$v - (e - 1) + (r - 1) = 2 \quad \Rightarrow \quad v - e + r = 2.$$

Thus, by induction, Euler's formula holds for every connected planar graph.

2019–20, Section C, Q1

Find the number of integers between 1 and 500 that are not divisible by any of 2, 3, 5 or 7.

Solution

We count integers from 1 to 500 not divisible by 2, 3, 5, 7. Use the **principle of inclusion–exclusion**.

Let

$$A_2 = \{n \leq 500 : 2 \mid n\}, \quad A_3 = \{n \leq 500 : 3 \mid n\}, \quad A_5, A_7 \text{ similarly.}$$

Step 1: Single divisors:

$$|A_2| = \left\lfloor \frac{500}{2} \right\rfloor = 250, \quad |A_3| = \left\lfloor \frac{500}{3} \right\rfloor = 166,$$

$$|A_5| = \left\lfloor \frac{500}{5} \right\rfloor = 100, \quad |A_7| = \left\lfloor \frac{500}{7} \right\rfloor = 71.$$

Sum $S_1 = 250 + 166 + 100 + 71 = 587$.

Step 2: Pairs (LCM):

$$|A_2 \cap A_3| = \left\lfloor \frac{500}{6} \right\rfloor = 83, \quad |A_2 \cap A_5| = \left\lfloor \frac{500}{10} \right\rfloor = 50,$$

$$|A_2 \cap A_7| = \left\lfloor \frac{500}{14} \right\rfloor = 35, \quad |A_3 \cap A_5| = \left\lfloor \frac{500}{15} \right\rfloor = 33,$$

$$|A_3 \cap A_7| = \left\lfloor \frac{500}{21} \right\rfloor = 23, \quad |A_5 \cap A_7| = \left\lfloor \frac{500}{35} \right\rfloor = 14.$$

Sum $S_2 = 83 + 50 + 35 + 33 + 23 + 14 = 238$.

Step 3: Triples:

$$|A_2 \cap A_3 \cap A_5| = \left\lfloor \frac{500}{30} \right\rfloor = 16, \quad |A_2 \cap A_3 \cap A_7| = \left\lfloor \frac{500}{42} \right\rfloor = 11,$$

$$|A_2 \cap A_5 \cap A_7| = \left\lfloor \frac{500}{70} \right\rfloor = 7, \quad |A_3 \cap A_5 \cap A_7| = \left\lfloor \frac{500}{105} \right\rfloor = 4.$$

Sum $S_3 = 16 + 11 + 7 + 4 = 38$.

Step 4: All four:

$$|A_2 \cap A_3 \cap A_5 \cap A_7| = \left\lfloor \frac{500}{210} \right\rfloor = 2.$$

So $S_4 = 2$.

Total divisible by at least one of 2, 3, 5, 7:

$$N_{\text{div}} = S_1 - S_2 + S_3 - S_4 = 587 - 238 + 38 - 2 = 385.$$

Total integers from 1 to 500 is 500, hence

$$N_{\text{not div}} = 500 - 385 = \boxed{115}.$$

2019–20, Section C, Q2

A collection of 10 electric bulbs contains 3 defective bulbs.

- (i) In how many ways can a sample of 4 bulbs be selected?
- (ii) In how many ways can a sample of 4 bulbs be selected which contains 2 good bulbs and 2 defective bulbs?
- (iii) In how many ways can a sample of 4 bulbs be selected so that either the sample contains 3 good bulbs and 1 defective bulb or 1 good bulb and 3 defective bulbs?

Solution

Let there be 7 good bulbs and 3 defective bulbs.

(i) Any sample of 4 bulbs:

$$\binom{10}{4} = 210.$$

(ii) **Exactly 2 good and 2 defective:** Choose 2 from the 7 good and 2 from the 3 defective:

$$\binom{7}{2} \binom{3}{2} = 21 \times 3 = 63.$$

(iii) **Either 3 good +1 defective or 1 good +3 defective:**

• 3 good and 1 defective:

$$\binom{7}{3} \binom{3}{1} = 35 \times 3 = 105.$$

• 1 good and 3 defective:

$$\binom{7}{1} \binom{3}{3} = 7 \times 1 = 7.$$

Total favourable samples:

$$105 + 7 = \boxed{112}.$$

3 III Semester Theory Examination 2020–21

2020–21, Section A, Q1

Define complete graph and regular graph.

Solution

Complete graph: Defined earlier: K_n is a simple graph with n vertices where every pair of distinct vertices is joined by an edge.

Regular graph: A graph G is called k -regular if every vertex of G has the same degree k . If a graph is k -regular for some k , we simply call it a *regular graph*.

Example: The cycle C_n is 2-regular; every vertex has degree 2.

2020–21, Section B, Q1

If a connected planar graph G has n vertices, e edges and r regions, prove that

$$n - e + r = 2.$$

Solution

This is the same Euler's formula for connected planar graphs, already proved in **2019–20, Section B, Q1**. The proof can be repeated here (trees as base case, then remove an edge on a cycle and use induction).

2020–21, Section C, Q1

Find the number of integers between 100 and 1000 that are divisible by 3 or 5 or 7.

Solution

We count integers from 100 to 1000 (inclusive) divisible by at least one of 3, 5, 7. First count up to 1000, then subtract numbers up to 99.

Up to 1000:

$$n_3 = \left\lfloor \frac{1000}{3} \right\rfloor = 333, \quad n_5 = \left\lfloor \frac{1000}{5} \right\rfloor = 200, \quad n_7 = \left\lfloor \frac{1000}{7} \right\rfloor = 142.$$

$$n_{3,5} = \left\lfloor \frac{1000}{15} \right\rfloor = 66, \quad n_{3,7} = \left\lfloor \frac{1000}{21} \right\rfloor = 47, \quad n_{5,7} = \left\lfloor \frac{1000}{35} \right\rfloor = 28.$$

$$n_{3,5,7} = \left\lfloor \frac{1000}{105} \right\rfloor = 9.$$

Using inclusion-exclusion:

$$N_{\leq 1000} = (333 + 200 + 142) - (66 + 47 + 28) + 9 = 675 - 141 + 9 = 543.$$

Up to 99:

$$m_3 = \left\lfloor \frac{99}{3} \right\rfloor = 33, \quad m_5 = \left\lfloor \frac{99}{5} \right\rfloor = 19, \quad m_7 = \left\lfloor \frac{99}{7} \right\rfloor = 14.$$

$$m_{3,5} = \left\lfloor \frac{99}{15} \right\rfloor = 6, \quad m_{3,7} = \left\lfloor \frac{99}{21} \right\rfloor = 4, \quad m_{5,7} = \left\lfloor \frac{99}{35} \right\rfloor = 2.$$

$$m_{3,5,7} = \left\lfloor \frac{99}{105} \right\rfloor = 0.$$

$$N_{\leq 99} = (33 + 19 + 14) - (6 + 4 + 2) + 0 = 66 - 12 = 54.$$

Between 100 and 1000:

$$N_{100 \text{ to } 1000} = N_{\leq 1000} - N_{\leq 99} = 543 - 54 = \boxed{489}.$$

4 III Semester Theory Examination 2021–22

2021–22, Section A, Q1

Explain Euler's formula. Determine the number of regions if a planar graph has 30 vertices of degree 3 each.

Solution

Euler's formula: For a connected planar graph,

$$v - e + r = 2.$$

We are given a planar graph with

$$v = 30, \quad \text{each vertex has degree 3.}$$

Step 1: Find the number of edges. By the Handshaking Lemma,

$$\sum \deg(v_i) = 2e.$$

Here each of the 30 vertices has degree 3, so

$$30 \times 3 = 2e \Rightarrow e = \frac{90}{2} = 45.$$

Step 2: Use Euler's formula to find number of regions r .

$$v - e + r = 2 \Rightarrow 30 - 45 + r = 2 \Rightarrow r = 2 + 15 = 17.$$

So the planar graph has $\boxed{17}$ regions.

2021–22, Section A, Q2

Explain Pigeonhole Principle with example.

Solution

Statement: If $n + 1$ objects are placed into n boxes, then at least one box contains at least two objects.

Example: In a group of 13 people, at least two must have their birthdays in the same month. (There are 12 months, so here $n = 12$. Putting 13 people into 12 “month boxes” forces one box to contain at least 2.)

2021–22, Section C, Q1

Explain the following terms with example:

1. Graph coloring and chromatic number.
2. How many edges in K_7 and $K_{3,3}$?
3. Isomorphic graph and Hamiltonian graph.
4. Bipartite graph.
5. Handshaking theorem.

Solution

(i) Graph coloring and chromatic number.

Graph coloring: A (vertex) *graph coloring* of a graph $G = (V, E)$ is an assignment of colors to the vertices of G such that adjacent vertices receive different colors. Formally, it is a function

$$c : V \rightarrow \{1, 2, \dots, k\}$$

for some positive integer k with the property that if $\{u, v\} \in E$ then $c(u) \neq c(v)$. Such a coloring is called a *proper vertex coloring*.

Chromatic number: The *chromatic number* of a graph G , denoted by $\chi(G)$, is the smallest positive integer k for which G admits a proper vertex coloring with k colors. *Example:* A cycle C_5 has chromatic number 3.

(ii) How many edges in K_7 and $K_{3,3}$?

For a complete graph K_n ,

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2},$$

so $|E(K_7)| = \binom{7}{2} = 21$.

For a complete bipartite graph $K_{m,n}$,

$$|E(K_{m,n})| = mn,$$

so $|E(K_{3,3})| = 3 \times 3 = 9$.

(iii) Isomorphic graph and Hamiltonian graph.

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if there exists a bijection $\varphi : V_1 \rightarrow V_2$ such that $\{u, v\} \in E_1 \iff \{\varphi(u), \varphi(v)\} \in E_2$.

A graph is *Hamiltonian* if it contains a cycle that visits every vertex exactly once (a Hamiltonian cycle).

(iv) Bipartite graph.

A graph $G = (V, E)$ is *bipartite* if V can be partitioned into two disjoint sets V_1 and V_2 such that every edge has one endpoint in V_1 and the other in V_2 . No edge joins two vertices in the same part.

(v) Handshaking theorem.

In any undirected graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

That is, the sum of the degrees of all vertices equals twice the number of edges. As a consequence, the number of vertices of odd degree in any graph is even.

2021–22, Section C, Q2

Explain the following with justification:

(i) In an undirected graph the total number of odd degree vertices is even.

(ii) The maximum number of edges in a simple graph on n vertices is $\frac{n(n-1)}{2}$.

Solution

(i) Number of odd degree vertices is even:

From the Handshaking Theorem,

$$\sum_{v \in V} \deg(v) = 2|E|,$$

which is an even integer.

Split the sum into degrees of odd-degree vertices and even-degree vertices. The sum of degrees of even-degree vertices is even. Therefore, the sum of degrees of odd-degree vertices must also be even.

The sum of an odd number of odd integers is odd; therefore, the number of odd-degree vertices cannot be odd. Hence it must be even.

(ii) Maximum number of edges in a simple graph on n vertices:

In a simple graph there is at most one edge between any pair of distinct vertices and no loops. Hence edges correspond to unordered pairs of distinct vertices.

The number of such pairs is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

This is achieved by the complete graph K_n , so the maximum possible number of edges is $\frac{n(n-1)}{2}$.

III Semester Theory Examination 2022–23

2022–23, Section A, Q1

Draw an adjacency matrix for the given graph.

Solution

The exact adjacency matrix depends on the specific graph provided in the question paper.

Method:

1. Label the vertices as $v_1, v_2, \dots, v_n$.
2. Form an $n \times n$ matrix $A = (a_{ij})$.
3. For each pair (i, j) , set $a_{ij} = 1$ if there is an edge between v_i and v_j , otherwise $a_{ij} = 0$.
4. For an undirected simple graph, A will be symmetric with all diagonal entries 0.

Students should fill in the 0-1 entries according to the particular graph shown in the exam.

2022–23, Section A, Q2

Describe planar graph and express Euler's formula for planar graph.

Solution

Planar graph: A graph that can be drawn in the plane without any edges crossing is called a *planar graph*. Such a crossing-free drawing is called a *plane graph*.

Euler's formula: For any connected planar graph with v vertices, e edges and r regions (faces),

$$v - e + r = 2.$$

For a planar graph with k connected components, this generalizes to

$$v - e + r = 1 + k.$$

2022–23, Section B, Q1

Explain the following:

1. Euler graph and Hamiltonian graph.
2. Chromatic number of a graph.
3. Walk and path.
4. Bipartite graph.

Solution

(i) Euler graph and Hamiltonian graph:

- An *Euler graph* is a connected graph containing an Eulerian circuit (a closed walk using every edge exactly once). All vertices have even degree.
- A *Hamiltonian graph* contains a Hamiltonian cycle that visits every vertex exactly once.

(ii) Chromatic number: Already defined: minimum number of colors required to color vertices so that adjacent vertices get different colors.

(iii) Walk and path:

- A *walk* is a finite sequence of vertices $v_0, v_1, \dots, v_k$ such that each consecutive pair is connected by an edge; vertices and edges may repeat.
- A *path* is a walk in which all vertices (and therefore edges) are distinct.

(iv) Bipartite graph: Defined earlier: vertex set can be partitioned into two sets V_1, V_2 with edges only between the parts. A graph is bipartite if and only if it has no odd cycle.

2022–23, Section C, Q4(b)

Using Pigeonhole Principle find the minimum number n of integers to be selected from

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

so that:

- the sum of two of the selected integers is even;
- the difference of two of the selected integers is 5.

Solution

(i) Sum of two selected integers is even.

An integer is either even or odd. In S there are

$$\text{Even: } \{2, 4, 6, 8\} \text{ (4 numbers),} \quad \text{Odd: } \{1, 3, 5, 7, 9\} \text{ (5 numbers).}$$

The sum of two integers is even if both are even or both are odd. We want the smallest n such that among the chosen integers there exist two of the same parity.

Worst case: choose numbers while trying to avoid this. We can pick at most one even and at most one odd (2 numbers) without forming an even-sum pair. On picking the third number, it must match the parity of one of the previous two, producing an even sum.

Hence, the minimum n is

$$\boxed{3}$$

(ii) The difference of two selected integers is 5.

We want the smallest n such that some chosen pair differs by 5.

Partition the set S into groups where numbers in the same group differ by 5:

$$\{1, 6\}, \{2, 7\}, \{3, 8\}, \{4, 9\}, \{5\}.$$

These 5 groups are the “boxes”. If we select at most one element from each group (i.e., 5 elements in total), we can avoid getting a pair with a difference of 5.

As soon as we pick the 6-th number, by the Pigeonhole Principle, two selected numbers will lie in the same group; hence their difference will be 5.

Therefore, the minimum such n is

$$\boxed{6}$$